

Homework 1 – Sessions 1 & 2

Algebra, order and justification · Sets, operations and double inclusion

Part A · Session 1 – algebra, order and justification

1 EXPONENTS

Simplify $\frac{16^{3/4} \cdot 2^{-3}}{8^{1/3}}$, naming the exponent law used at each step.

2 MISSING HYPOTHESIS

Each line below is wrong as written. For each, give a counterexample, then either add a hypothesis that makes it true or explain why no such repair exists.

a. $\sqrt{ab} = \sqrt{a} \sqrt{b}$

b. $\frac{1}{a+b} = \frac{1}{a} + \frac{1}{b}$

3 ABSOLUTE VALUE

Write “ x is at least 5 away from -2 ” as an absolute-value inequality and solve it. Explain why the solution set splits into two pieces.

4 COMPLETING THE SQUARE

Write $2x^2 - 12x + 23$ in the form $a(x - h)^2 + k$.

5 RATIONAL INEQUALITY

Find all real x with $\frac{2}{x-1} \leq 1$. Handle the sign of $x - 1$ explicitly; do not multiply through by an expression whose sign you do not know.

6 QUADRATIC REASONING

Solve

$$2x^2 - 8x + 3 = 0$$

by completing the square. Do not use the quadratic formula.

- Calculate the discriminant and state what it tells you about the number of real solutions.
- Find the roots by completing the square, writing each line as an equivalence.

7 $\in$ VS $\subseteq$

Let $C = \{1, \{1\}, \{1, 2\}, \emptyset\}$. Decide true or false, and in each case quote the definition that settles it: $1 \in C$, $\{1\} \subseteq C$, $\{1, 2\} \subseteq C$, $\emptyset \subseteq C$.

8 OPERATIONS

Let $U = \{1, 2, \dots, 15\}$, $A = \{x \in U : x \text{ is a multiple of } 3\}$ and $B = \{x \in U : x \text{ is odd}\}$. Write A and B in roster notation, compute $A \triangle B$ and $(A \cup B)^c$, then describe $(A \cup B)^c$ in plain English and name the law that turns “not (a multiple of 3 or odd)” into a statement with “and” in it.

9 POWER SET

- With $A = \{a, b\}$ and $B = \{b, c\}$, list $\mathcal{P}(A \cap B)$ and $\mathcal{P}(A) \cap \mathcal{P}(B)$ in full, and check they agree.
- Prove that $\mathcal{P}(A \cap B) = \mathcal{P}(A) \cap \mathcal{P}(B)$ for *all* sets A, B , by double inclusion.

10 DE MORGAN

Let $A, B \subseteq U$. Prove that $(A \cap B)^c = A^c \cup B^c$, by double inclusion. The word **arbitrary** must appear in each direction.

11 DISPROVE, THEN REPAIR

- Disprove: $(A \cup B) \setminus B = A$ for all sets A, B . Name your sets, compute both sides, and point at an element that separates them.
- State the extra condition on A and B that makes the identity true, and prove it under that condition.

12 APPLICATION · CAMPUS DATA

A campus card system records, for one week, which of the **200** students used the gym and which used the library. Take U to be the set of all **200** students, $G \subseteq U$ those who used the gym and $L \subseteq U$ those who used the library. The logs give

$$|G| = 120, \quad |L| = 95, \quad |G \cap L| = 60.$$

- The two sets cut U into four regions. Name each one in set notation and give its size, explaining where each number comes from.
- Every visit is logged as an ordered pair (student, facility), the facility drawn from $F = \{\text{gym, library, pool}\}$. How many such pairs are possible in principle? Name the theorem that gives the count, and say why the pairs must be ordered.